

ASU Macro Qual Module 6

Galina Closed-Form and Creative Dynamic Problems Lab

Exhaustive but Exam-Facing Notes for Reading, Formulation, Recursive Arguments, Economic Intuition, and Closed-Form Derivations

Built from Galina notes, Galina homeworks, current PS1, and ASU Macro quals

May 15, 2026

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How to use this lab

This packet is a problem-solving lab, not a standard lecture summary. The goal is to make creative dynamic problems feel routine. Every worked pattern follows the same exam sequence:

1. identify the state variable and timing;
2. write the Bellman or sequential problem;
3. derive the FOC, envelope, Euler equation, or cutoff condition;
4. guess a functional form only after seeing homogeneity, quadratic structure, or stationarity;
5. match coefficients and verify feasibility, transversality, uniqueness, and intuition.

The target weaknesses are reading comprehension, formulation and setup, recursive arguments, economic intuition, and closed-form derivations. The packet therefore deliberately repeats a few core moves in many disguises.

1 Source inventory

1.1 Accessible source base and extraction notes

Source	Type	Topics used	Extraction and reliability notes
MACRO_MANIFEST.md	Project manifest	Locked Module 6 scope, source hierarchy, required packet outputs, notation conventions, required worked examples.	Direct markdown; treated as controlling instructions for structure and coverage.
SKILL.md	PhD Qual Lecture Notes Builder skill	Required workflow: source inventory, pattern analysis, blueprint, packet, exercise bank, answer sketches, quality audit.	Direct markdown; used as the packet-production standard.
GitHub folder Macroeconomics/ Galina/	Repository source folder	Confirmed Galina folder contents: class note folders, homeworks, PS1-5, and a dynamic programming PDF.	Web folder was accessible; raw lecture TeX exists in the repository, but uploaded PDFs and homeworks were used as the main mathematical source because the user supplied them.
Class01_OGM_T.pdf	Professor handwritten lecture note	Finite-horizon OGM, feasibility, KKT, terminal capital, Euler equation, second-order difference equation.	Scanned handwritten notes; OCR imperfect. Equations were cross-checked against HW1 and HW2.
Class02OGM, Tsolution(1).pdf	Professor handwritten lecture note	Shooting algorithm; finite-horizon log-linear solution; saving-rate recursion; consumption path.	Scanned handwritten notes; formulas were inferred and checked against HW2.
Class03OGM, infinitehorizon.pdf	Professor handwritten lecture note	Infinite-horizon sequential problem, Euler equation, transversality condition, sufficiency/necessity logic.	Scanned notes; transversality and Euler notation cleaned for readability.
Class04GuessandVerify, SS, Recursiveformulation. pdf	Professor handwritten lecture note	Guess-and-verify; log Cobb-Douglas; steady state; recursive formulation and principle of optimality.	Scanned notes; central formulas reconstructed from visible pages and homeworks.
Class05DP,CMT,BSC, BE.pdf	Professor handwritten lecture note	Bellman operator, contraction mapping theorem, Blackwell sufficient conditions.	Scanned notes; used only for the reusable operator templates and not for fine typography.

Source	Type	Topics used	Extraction and reliability notes
Class06SolvingRPaandcharPactesrizingVLpdf	Professor hand-written lecture note	Existence/solution of recursive problems and value function characterization.	OCR was limited; used in broad role only.
Class07concavityFOCEE.Professor hand-written lecture note	Professor hand-written lecture note	Monotonicity, concavity, FOC, envelope condition, Euler equation from recursive problem.	Scanned notes; formulas cross-checked against standard Bellman derivations.
Class08-numericalVFIetC.Professor hand-written lecture note	Professor hand-written lecture note	Value function iteration, grid bounds, policy-function computation, numerical cautions.	Included only as a trap/-diagnostic source.
HW1-Galina.pdf	Homework	Multi-stage seeding/harvesting/processing model; KKT; terminal conditions; shooting algorithm for finite OGM.	Text extraction good; used for setup drills and KKT timing.
HW2-Galina.pdf	Homework	Finite-horizon log-linear and log Cobb-Douglas OGM closed forms; limiting infinite-horizon path.	Text extraction good; used for finite-horizon saving-rate recursions.
HW3-Galina.pdf	Homework	Cake-eating CRRA; infinite OGM with depreciation; recursive formulation; exogenous income state.	Text extraction good; used for cake-eating and recursive-state drills.
HW4-Galina.pdf	Homework	Blackwell fixed point; endogenous population growth AK; recursive comparative statics; skill accumulation.	Text extraction good; skill problem matches the 2021 prelim.
HW5-Galina.pdf	Homework	VFI; fluctuating fixed income; disaster-risk AK; recursive setup variants including labor adjustment and costly adoption.	Text extraction good; disaster problem matches the 2024 qualifier.
PS1-5.pdf / PS1: DynamicProgramming	Current 2026 problem set	Reaching the center; tree cutting/stopping; growth with leisure; stochastic growth and bounded state restriction.	Text extraction good; used as the current-course anchor for Module 6.

Source	Type	Topics used	Extraction and reliability notes
2016-06...tex– 2024-06...tex	Past ASU Macro quals	Natural resource extraction (2018), irreversible work choice (2019), public leisure/growth (2020), skill accumulation (2021), project choice with survival (2022), alternating management rights (2023), disaster-risk AK (2024), plus recurring recursive and closed-form prompts.	TeX files were readable; exact question numbering in the original files is sometimes distorted by conversion, so the practice map labels them by year and topic.

1.2 Source hierarchy implemented

The source hierarchy used here is:

1. Past ASU Macro qualifiers set the exam archetypes and practice map.
2. Current PS1 sets current-course emphasis: reaching-center, tree cutting, abstract Bellman mapping, bounded stochastic growth.
3. Galina lecture notes set notation and core OGM/DP derivation style.
4. Galina homeworks supply practice depth and many closed-form mutations.

2 Detailed blueprint

2.1 Module objective

By the end of this lab, a student should be able to read a creative dynamic problem, identify the state and timing, write the correct recursive or sequential formulation, derive the Euler equation or cutoff rule, and exploit log/CRRA/quadratic/homogeneous structure for closed-form policies.

2.2 Prerequisite concepts

- Static constrained optimization and Kuhn-Tucker conditions.
- Concavity and why FOCs are sufficient under concavity.
- Bellman equations, feasible correspondences, value functions, policy functions.
- Envelope condition and Euler equation.
- Transversality and no-waste terminal conditions.
- Homogeneity and coefficient matching.

2.3 Canonical patterns

Pattern	Recognition clues	Core technique	Typical closed form
Quadratic adjustment/reaching	Squared distance today plus discounted future squared distance.	Guess $v(x) = Ax^2$; minimize quadratic; solve scalar fixed point.	Linear law $x' = \lambda x$.
Stopping/tree cutting	Option to stop now or continue one deterministic transition.	Bellman as $v(x) = \max\{\text{stop}, \text{continue}\}$; contraction; cutoff proof.	Cut when $x \geq x^*$.
Cake/resource eating	Fixed stock, choose extraction/consumption over time.	Euler plus resource PV or guess $V(w) = Bw^{1-\sigma}/(1-\sigma)$.	Consume constant fraction.
Finite OGM	Terminal date, no value after T , no idle capital.	KKT; terminal condition; backward saving recursion.	s_t depends on time-to-go.
Log Cobb-Douglas OGM	$u = \log c$, $f(k) = Ak^\alpha$.	Guess constant saving rate or $V = a + b \log k$.	$k' = \alpha\beta Ak^\alpha$ in infinite horizon.
AK log model	Linear technology, log utility.	Homogeneity; coefficient matching.	$k' = \beta Ak$, $c = (1 - \beta)Ak$.
Natural resource	Unused stock grows; extraction produces consumption.	State is resource stock; Euler for extracting now versus leaving stock to grow.	$l_t = (1 - \beta)[\beta(1 + g)]^t x_0$.
Disaster-risk AK	Linear production with multiplicative capital survival shock.	Stochastic Bellman; CRRA homogeneity.	Saving share $s = [\beta A^{1-\sigma} H]^{1/\sigma}$.
Human capital investment	Current income finances consumption and investment; skill law homogeneous in skill and investment.	Guess $V = a + B \log s$ and $i = \eta s$.	Linear investment rule $i(s) = \eta s$.
Project choice/cut-off	Discrete project choice plus continuous saving; failure state.	Conditional value functions; compare affine functions of $\log k$.	Safe project above a wealth cutoff.
Alternating control	Two-period manager problem, current manager values own current and next consumption.	Guess symmetric linear rule; match $V = a + b \log k$.	Joint asset grows by $\beta^2 A$ instead of βA .
Multi-stage production	State has pipeline variables or delayed outputs.	Careful timing; KKT by calendar date.	Euler equates processing today to seeding for tomorrow.

2.4 Exam variations to expect

- The problem will add one state variable: age, eligibility, tax state, disaster state, previous sector, population, or project status.
- The problem will hide a one-period timing lag: seeds planted today harvest tomorrow, accepted

job starts next period, project fails before output, withdrawn asset grows only one period.

- The problem will ask for both formal and intuitive responses. The formal answer is usually one FOC, one envelope, one Euler, or one cutoff inequality. The intuition is the marginal tradeoff.
- The problem will include an Economist X claim. Convert the claim into a sign statement and check it against the policy formula or recursive monotonicity argument.

2.5 Closed-form catalog to master

1. Quadratic reaching: $v(x) = Ax^2$, $x' = x/(2 + \beta A)$.
2. Cake eating CRRA: $w' = \beta^{1/\sigma} w$, $c = (1 - \beta^{1/\sigma})w$.
3. Finite log-linear OGM: $c_t = (\beta R)^t c_0$, $c_0 = \frac{1-\beta}{1-\beta^{T+1}} R k_0$.
4. Finite log Cobb-Douglas: $s_t = \frac{\alpha\beta(1-(\alpha\beta)^{T-t})}{1-(\alpha\beta)^{T-t+1}}$.
5. Infinite log Cobb-Douglas: $k' = \alpha\beta A k^\alpha$.
6. AK log: $k' = \beta A k$, $c = (1 - \beta) A k$.
7. Natural resource log/power: $l_t = (1 - \beta)[\beta(1 + g)]^t x_0$, $c_t = l_t^\gamma$.
8. Endogenous population AK: $c = (1 - \beta) A k$, long-run gross population growth = βA .
9. Disaster-risk AK CRRA: $s = [\beta A^{1-\sigma}((1 - q) + q\rho^{1-\sigma})]^{1/\sigma}$.
10. Human capital: $i = \eta s$ where η solves a one-dimensional equation.
11. Project cutoff: compare $a_i + b_i \log k$ across projects.
12. Alternating management: symmetric joint saving $k' = \beta^2 A k$.

2.6 Suggested study plan

- **Block 1, 90 minutes:** read Sections 3–4; redo the derivation templates from memory.
- **Block 2, 2 hours:** work Sections 5–6; derive every closed form without looking.
- **Block 3, 2 hours:** work Sections 7 and 11; focus on state choice and cutoff arguments.
- **Block 4, 60 minutes:** use the final checklist and practice map to select old qual problems.

3 Pattern recognition guide

3.1 The first 90 seconds of any creative dynamic problem

When you open a problem, do not start with algebra. First answer these questions in the margin.

1. **What is predetermined at the start of the period?** This is the state.
2. **What is chosen after the state is observed?** These are controls.
3. **What current payoff is received?** This pins down the timing.
4. **What maps into next period's state?** This is the transition law.
5. **Is the problem stationary?** If yes, use one Bellman equation; if no, index values by time, age, status, or state.
6. **Is there a homogeneity clue?** Log utility plus Cobb-Douglas or AK technology usually implies $V = a + b \log x$ and a constant saving share.
7. **Is there a discrete choice?** Solve conditional value functions first, then compare them.

Exam tip

A good recursive formulation is often worth more than a long but mistimed derivation. In Galina-style creative questions, the most common loss of points is using the right Euler equation with the wrong state variable.

3.2 State-variable dictionary

Problem language	State variable	Why
Cake/resource stock at beginning of period	Remaining stock w or x	It constrains current consumption/extraction and becomes next stock after the choice.
Capital available before production	Capital k	Output $f(k)$ is immediately available for c and k' .
Capital saved may be destroyed before next production	Current surviving capital k and shock transition for k'	The marginal value of saving is expected and state-contingent.
Tree grows if not cut	Tree size x	Stop value and continuation value depend only on current tree size.
Income follows $y' = h(y)$	Pair (k, y) or sufficient lags	Income is payoff-relevant and predicts future income.
Alternating management	Joint asset k plus manager identity, or two-period Bellman by manager	The current manager changes the objective and next controller.
Project fails and insurance begins	Status $a \in \{\text{alive, failed}\}$ plus capital/resources	Once failed, the continuation value is an absorbing constant.
Discrete technology adoption	Capital k and adoption status $a \in \{L, H\}$	Once adopted, high productivity is permanent.

Problem language	State variable	Why
Labor adjustment cost $C(n, n')$	Capital k and current labor n	Tomorrow's labor choice affects current adjustment cost.

3.3 Guess dictionary

Reusable formula

Structure	Guess
Quadratic loss and linear transition	$v(x) = Ax^2$; policy linear.
Log utility, linear technology/resource transition	$V(k) = a + b \log k$; policy is a constant fraction.
Log utility, Cobb-Douglas Ak^α	$V(k) = a + b \log k$; policy $k' = sAk^\alpha$.
CRRA utility, linear homogeneous transition	$V(k) = Bk^{1-\sigma}/(1-\sigma) + D$; policy constant fraction.
Discrete choice with log-homogeneous conditional problems	$V_i(k) = a_i + b_i \log k$; compare lines in $\log k$.
Stopping problem	$v(x) = \max\{\text{stop payoff}, \text{continue payoff}\}$; cutoff if single crossing.

4 Core derivation templates

4.1 Template 1: finite-horizon KKT for OGM

Consider

$$\begin{aligned} \max_{\{c_t, k_{t+1}\}_{t=0}^T} \quad & \sum_{t=0}^T \beta^t u(c_t) \\ \text{s.t.} \quad & c_t + k_{t+1} \leq f(k_t), \quad k_{t+1} \geq 0, \quad k_0 \text{ given.} \end{aligned}$$

The Lagrangian can be written as

$$\mathcal{L} = \sum_{t=0}^T \beta^t [u(c_t) - \lambda_t (c_t + k_{t+1} - f(k_t)) + \mu_t k_{t+1}].$$

The FOCs are

$$\begin{aligned} (c_t) : \quad & u'(c_t) - \lambda_t = 0, \quad t = 0, \dots, T, \\ (k_{t+1}) : \quad & -\lambda_t + \mu_t + \beta \lambda_{t+1} f'(k_{t+1}) = 0, \quad t = 0, \dots, T-1, \\ (k_{T+1}) : \quad & -\lambda_T + \mu_T = 0. \end{aligned}$$

Complementary slackness is

$$\begin{aligned} \lambda_t [c_t + k_{t+1} - f(k_t)] &= 0, \quad \lambda_t \geq 0, \\ \mu_t k_{t+1} &= 0, \quad \mu_t \geq 0. \end{aligned}$$

Under Inada utility, $c_t > 0$, so $\lambda_t = u'(c_t) > 0$ and the resource constraint binds. At the terminal date, $\lambda_T > 0$, hence $\mu_T > 0$, so $k_{T+1} = 0$.

For interior $k_{t+1} > 0$ with $t < T$, $\mu_t = 0$ and

$$u'(c_t) = \beta u'(c_{t+1}) f'(k_{t+1}).$$

Using $c_t = f(k_t) - k_{t+1}$ gives the second-order difference equation

$$u'(f(k_t) - k_{t+1}) = \beta f'(k_{t+1}) u'(f(k_{t+1}) - k_{t+2}), \quad k_0 \text{ given}, \quad k_{T+1} = 0.$$

Common trap

Do not impose $k_T = 0$ in the standard one-stage finite OGM. The terminal no-waste condition is $k_{T+1} = 0$. In a delayed production pipeline, the last useful state may be earlier; see Section 7.7.

4.2 Template 2: Bellman to Euler

For a stationary deterministic problem

$$V(k) = \max_{k' \in \Gamma(k)} \{F(k, k') + \beta V(k')\},$$

with interior policy $g(k)$, the FOC and envelope condition are

$$\begin{aligned} F_2(k, g(k)) + \beta V'(g(k)) &= 0, \\ V'(k) &= F_1(k, g(k)). \end{aligned}$$

Evaluating the envelope condition one period ahead gives the Euler equation

$$F_2(k_t, k_{t+1}) + \beta F_1(k_{t+1}, k_{t+2}) = 0.$$

For OGM, $F(k, k') = u(f(k) - k')$, so

$$u'(f(k_t) - k_{t+1}) = \beta f'(k_{t+1}) u'(f(k_{t+1}) - k_{t+2}).$$

4.3 Template 3: coefficient matching with $V = a + b \log x$

Suppose

$$V(x) = \max_{0 \leq y \leq Ax} \{\log(Ax - y) + \beta V(y)\}.$$

Guess $V(x) = a + b \log x$. The FOC is

$$\frac{1}{Ax - y} = \frac{\beta b}{y} \quad \Rightarrow \quad y = \frac{\beta b}{1 + \beta b} Ax.$$

Substitute the policy into the Bellman equation. The coefficient on $\log x$ must satisfy

$$b = 1 + \beta b \quad \Rightarrow \quad b = \frac{1}{1 - \beta}.$$

Then $y = \beta Ax$ and $c = (1 - \beta)Ax$.

Exam tip

When the coefficient on $\log x$ is b , do not set $b = 1/(1 - \beta)$ automatically. With Cobb-Douglas Ak^α , the coefficient is $b = \alpha/(1 - \alpha\beta)$. With endogenous population and $k' = \text{constant} \cdot k^{1-\gamma}$, it is $b = 1/[1 - \beta(1 - \gamma)]$.

4.4 Template 4: discrete choice by conditional value functions

If the agent chooses project $i \in \{S, R\}$, solve the continuous problem conditional on i :

$$V_i(k) = \max_{a'} \{\text{current payoff under } i + \beta \mathbb{E}[V_i(k') \mid i]\}.$$

Then compare

$$V_S(k) - V_R(k).$$

If $V_i(k) = a_i + b_i \log k$, then

$$V_S(k) - V_R(k) = (a_S - a_R) + (b_S - b_R) \log k.$$

A cutoff exists when $b_S - b_R$ has a definite sign. If $b_S > b_R$, choose S for all

$$k \geq \bar{k} \equiv \exp\left(-\frac{a_S - a_R}{b_S - b_R}\right),$$

with the obvious boundary conventions if the right-hand side is outside the feasible range.

4.5 Template 5: stopping problems

A stopping problem usually has the form

$$V(x) = \max\{S(x), C(x) + \beta V(h(x))\}.$$

The proof of a cutoff has three parts:

1. show the Bellman operator is a contraction or monotone iteration is well behaved;
2. show the stop and continuation values cross once;
3. translate the crossing into a policy: stop above or below the threshold.

5 Worked examples: quadratic, stopping, and resource allocation

5.1 Worked example 1: reaching the center

Recognition clues. The problem has squared movement cost plus squared distance-to-target cost. This screams quadratic value function and linear policy.

Static problem. Given $x_0 \in [-1, 1]$, choose $x \in [-1, 1]$ to minimize

$$(x - x_0)^2 + x^2.$$

The FOC is

$$2(x - x_0) + 2x = 0 \quad \Rightarrow \quad x^* = \frac{x_0}{2}.$$

Since $x_0/2 \in [-1, 1]$, the constraint is not binding.

Finite horizon. Let $v_n(x)$ be the minimized discounted cost with n periods left. Use the recursion

$$v_n(x) = \min_{y \in [-1, 1]} \{(y - x)^2 + y^2 + \beta v_{n-1}(y)\},$$

with $v_0(x) = 0$ if no periods remain. Guess $v_n(x) = A_n x^2$. Then

$$v_n(x) = \min_y \{(y - x)^2 + (1 + \beta A_{n-1})y^2\}.$$

The minimizer is

$$y_n(x) = \frac{x}{2 + \beta A_{n-1}},$$

and the coefficient recursion is

$$A_n = \frac{1 + \beta A_{n-1}}{2 + \beta A_{n-1}}.$$

This proves by induction that $v_n(x) = A_n x^2$.

Infinite horizon. Guess $v(x) = Ax^2$ with $A \geq 0$:

$$A = \frac{1 + \beta A}{2 + \beta A}.$$

Equivalently,

$$\beta A^2 + (2 - \beta)A - 1 = 0.$$

The unique nonnegative solution is

$$A^* = \frac{\beta - 2 + \sqrt{\beta^2 + 4}}{2\beta}.$$

The optimal law of motion is

$$x_{t+1} = \lambda x_t, \quad \lambda = \frac{1}{2 + \beta A^*} = 1 - A^*.$$

Verification checklist

For quadratic guesses, verify: (i) the guessed coefficient is nonnegative; (ii) the policy respects bounds; (iii) the fixed point has the unique admissible root; (iv) the policy moves the state toward the target.

5.2 Worked example 2: tree cutting as stopping

Recognition clues. The agent chooses when to stop. The state grows deterministically if the agent waits. The payoff is realized only when cutting.

Bellman equation. Normalize the wood price to one. State x is tree size. The recursive problem is

$$V(x) = \max\{x, \beta V(h(x))\}.$$

The Bellman operator is

$$(TV)(x) = \max\{x, \beta V(h(x))\}.$$

For any two bounded functions V, W ,

$$|(TV)(x) - (TW)(x)| \leq \beta \|V - W\|_\infty,$$

so T is a contraction under the sup norm on the appropriate bounded/anchored function space.

Cutoff logic. Suppose there is an x^* such that

$$\frac{\beta h(x)}{x} > 1 \quad \text{if } x < x^*, \quad \frac{\beta h(x)}{x} < 1 \quad \text{if } x > x^*.$$

Then below x^* the one-period growth benefit is strong enough to justify waiting, while above x^* discounting dominates growth. The optimal rule is to cut once the tree is at least x^* .

Common trap

Do not write a differentiable FOC for the cutting date unless the problem explicitly asks for a continuous-time or continuous-date approximation. In discrete stopping, the primitive object is the max between stop and continue values.

5.3 Worked example 3: cake eating with CRRA utility

Environment. The consumer has cake w_t at the start of period t and chooses c_t and w_{t+1} :

$$c_t + w_{t+1} \leq w_t, \quad w_{t+1} \geq 0.$$

Preferences are

$$\sum_{t=0}^{\infty} \beta^t \frac{c_t^{1-\sigma} - 1}{1-\sigma}, \quad \sigma > 0,$$

with the log case obtained as $\sigma = 1$.

Bellman equation.

$$V(w) = \max_{0 \leq w' \leq w} \{u(w - w') + \beta V(w')\}.$$

The Euler equation is

$$u'(c_t) = \beta u'(c_{t+1}).$$

For CRRA, $u'(c) = c^{-\sigma}$, so

$$c_{t+1} = \beta^{1/\sigma} c_t.$$

Let $q = \beta^{1/\sigma}$. Since the stock is exhausted only asymptotically,

$$w_0 = \sum_{t=0}^{\infty} c_t = c_0 \sum_{t=0}^{\infty} q^t = \frac{c_0}{1 - q}.$$

Therefore

$$c_t = (1 - q)q^t w_0, \quad w_{t+1} = q w_t, \quad c(w) = (1 - q)w.$$

For log utility, $q = \beta$, so $c = (1 - \beta)w$ and $w' = \beta w$.

Steady state. A positive steady state would require $w' = w$, hence $c = 0$. Inada utility rules that out. The only steady state is the limiting boundary $w = 0$.

Exam tip

Cake eating is the no-growth benchmark behind many AK and resource problems. Once a stock grows at gross rate R , replace the Euler ratio by $c_{t+1}/c_t = (\beta R)^{1/\sigma}$, then use the correct present-value resource constraint.

6 Worked examples: optimal growth closed forms

6.1 Worked example 4: finite-horizon log utility and linear production

Environment. Let $u(c) = \log c$ and $f(k) = Rk$. The finite-horizon problem is

$$\begin{aligned} \max_{\{c_t, k_{t+1}\}_{t=0}^T} \quad & \sum_{t=0}^T \beta^t \log c_t \\ \text{s.t.} \quad & c_t + k_{t+1} = Rk_t, \quad k_0 \text{ given}, \quad k_{T+1} = 0. \end{aligned}$$

Euler equation. Since $u'(c) = 1/c$ and $f'(k) = R$,

$$\frac{1}{c_t} = \beta R \frac{1}{c_{t+1}} \Rightarrow c_{t+1} = \beta R c_t.$$

Lifetime resource constraint. Iterating the budget constraints gives

$$c_0 + \frac{c_1}{R} + \cdots + \frac{c_T}{R^T} = Rk_0.$$

Substitute $c_t = (\beta R)^t c_0$:

$$c_0 \sum_{t=0}^T \beta^t = Rk_0 \quad \Rightarrow \quad c_0 = \frac{1 - \beta}{1 - \beta^{T+1}} Rk_0.$$

Thus

$$c_t = (\beta R)^t \frac{1 - \beta}{1 - \beta^{T+1}} Rk_0.$$

Saving-rate recursion. Define

$$s_t \equiv \frac{k_{t+1}}{Rk_t}.$$

Then $c_t = (1 - s_t)Rk_t$ and $k_{t+1} = s_t Rk_t$. The Euler equation implies

$$s_t = \frac{\beta(1 - \beta^{T-t})}{1 - \beta^{T-t+1}}, \quad t = 0, \dots, T,$$

with $s_T = 0$.

6.2 Worked example 5: finite-horizon log utility and Cobb-Douglas production

Environment. Let $u(c) = \log c$ and $f(k) = Ak^\alpha$, $\alpha \in (0, 1)$:

$$c_t + k_{t+1} = Ak_t^\alpha, \quad k_{T+1} = 0.$$

Define the saving rate

$$s_t \equiv \frac{k_{t+1}}{Ak_t^\alpha}.$$

Then

$$c_t = (1 - s_t)Ak_t^\alpha, \quad k_{t+1} = s_t Ak_t^\alpha.$$

Euler equation.

$$\frac{1}{c_t} = \beta \alpha A k_{t+1}^{\alpha-1} \frac{1}{c_{t+1}}.$$

Substitute the saving-rate expressions:

$$\frac{s_t}{(1 - s_t)k_{t+1}} = \frac{\alpha \beta}{(1 - s_{t+1})k_{t+1}}.$$

Therefore

$$s_t(1 - s_{t+1}) = \alpha \beta (1 - s_t).$$

With terminal condition $s_T = 0$, the backward solution is

$$s_t = \frac{\alpha \beta [1 - (\alpha \beta)^{T-t}]}{1 - (\alpha \beta)^{T-t+1}}, \quad t = 0, \dots, T.$$

As $T \rightarrow \infty$ for fixed t , $s_t \rightarrow \alpha \beta$.

6.3 Worked example 6: infinite-horizon log Cobb-Douglas growth

Bellman equation.

$$V(k) = \max_{0 \leq k' \leq Ak^\alpha} \{\log(Ak^\alpha - k') + \beta V(k')\}.$$

Guess

$$V(k) = a + b \log k.$$

The FOC is

$$\frac{1}{Ak^\alpha - k'} = \frac{\beta b}{k'} \Rightarrow k' = \frac{\beta b}{1 + \beta b} Ak^\alpha.$$

The coefficient on $\log k$ after substitution is $\alpha + \alpha\beta b$, so

$$b = \alpha + \alpha\beta b \Rightarrow b = \frac{\alpha}{1 - \alpha\beta}.$$

Then

$$\frac{\beta b}{1 + \beta b} = \alpha\beta,$$

so the optimal policy is

$$k' = \alpha\beta Ak^\alpha, \quad c = (1 - \alpha\beta)Ak^\alpha.$$

Steady state. A steady state satisfies $k' = k = k^*$:

$$k^* = \alpha\beta A(k^*)^\alpha \Rightarrow k^* = (\alpha\beta A)^{1/(1-\alpha)}.$$

6.4 Worked example 7: AK log growth

Let $f(k) = Ak$ and $u(c) = \log c$. The Bellman equation is

$$V(k) = \max_{0 \leq k' \leq Ak} \{\log(Ak - k') + \beta V(k')\}.$$

Guess $V(k) = a + b \log k$. The FOC gives

$$k' = \frac{\beta b}{1 + \beta b} Ak.$$

Coefficient matching gives $b = 1 + \beta b$, hence $b = 1/(1 - \beta)$. Therefore

$$k' = \beta Ak, \quad c = (1 - \beta)Ak.$$

The gross growth factor of capital and consumption is βA .

Common trap

The policy $k' = \beta Ak$ does not mean the saving share out of resources is βA . The saving share out of output Ak is β ; the growth rate of the state is βA .

6.5 Worked example 8: finite horizon versus infinite horizon intuition

In the finite log Cobb-Douglas model,

$$s_t = \frac{\alpha\beta[1 - (\alpha\beta)^{T-t}]}{1 - (\alpha\beta)^{T-t+1}}.$$

For any fixed remaining horizon, $s_T = 0$ in the last period because there is no future. As the horizon grows, the early saving rate approaches the infinite-horizon constant $\alpha\beta$. This is a frequent exam intuition question: the infinite-horizon path is the limit of the finite-horizon solution away from the terminal date.

7 Worked examples: creative Galina-style dynamic problems**7.1 Worked example 9: natural resource extraction**

Environment. The state is a resource stock x_t . The agent extracts $l_t \in [0, x_t]$, produces $c_t = f(l_t)$, and the unextracted resource grows:

$$x_{t+1} = (x_t - l_t)(1 + g).$$

Sequential problem.

$$\begin{aligned} \max_{\{l_t\}_{t=0}^{\infty}} \quad & \sum_{t=0}^{\infty} \beta^t u(f(l_t)) \\ \text{s.t.} \quad & 0 \leq l_t \leq x_t, \quad x_{t+1} = (x_t - l_t)(1 + g), \quad x_0 > 0. \end{aligned}$$

Bellman equation. Using x' as the control,

$$V(x) = \max_{0 \leq x' \leq (1+g)x} \left\{ u \left(f \left(x - \frac{x'}{1+g} \right) \right) + \beta V(x') \right\}.$$

Euler equation. Let $l_t = x_t - x_{t+1}/(1+g)$. The FOC and envelope imply

$$u'(f(l_t))f'(l_t) = \beta(1+g)u'(f(l_{t+1}))f'(l_{t+1}).$$

The left side is the marginal utility value of extracting one more unit now. The right side is the discounted value of leaving it in the ground, where it becomes $1+g$ units of future resource.

Steady state condition. If an interior steady state has constant extraction, the Euler equation requires

$$1 = \beta(1+g).$$

A positive interior steady state is therefore knife-edge in this simple resource model.

Closed form with $u(c) = \log c$ **and** $f(l) = l^\gamma$. Then $u(f(l)) = \gamma \log l$. The Euler equation becomes

$$\frac{\gamma}{l_t} = \beta(1+g) \frac{\gamma}{l_{t+1}} \Rightarrow l_{t+1} = \beta(1+g)l_t.$$

The stock law implies the present-value resource identity

$$x_0 = \sum_{t=0}^{\infty} \frac{l_t}{(1+g)^t}.$$

Using $l_t = [\beta(1+g)]^t l_0$,

$$x_0 = l_0 \sum_{t=0}^{\infty} \beta^t = \frac{l_0}{1-\beta}.$$

Therefore

$$l_t = (1-\beta)[\beta(1+g)]^t x_0, \quad c_t = l_t^\gamma.$$

Consumption is constant iff $\beta(1+g) = 1$.

7.2 Worked example 10: disaster-risk AK model

Recognition clues. Linear production plus CRRA utility plus multiplicative shock means a constant saving share. The disaster changes the expected marginal continuation value.

Timing and Bellman equation. Let k be the surviving capital available at the start of a period. Output is Ak . The planner chooses next-period asset $k' \in [0, Ak]$. At the start of next period, the asset survives fully with probability $1-q$ and is reduced to $\rho k'$ with probability q . The Bellman equation is

$$V(k) = \max_{0 \leq k' \leq Ak} \left\{ u(Ak - k') + \beta \left[(1-q)V(k') + qV(\rho k') \right] \right\}.$$

Euler equation. The FOC is

$$u'(Ak - k') = \beta \left[(1-q)V'(k') + q\rho V'(\rho k') \right].$$

The envelope condition is

$$V'(k) = Au'(Ak - g(k)).$$

Thus, along the optimal path,

$$u'(c_t) = \beta \text{AE} \left[z_{t+1} u'(c_{t+1}(z_{t+1})) \right],$$

where $z_{t+1} = 1$ with probability $1-q$ and $z_{t+1} = \rho$ with probability q .

Closed form under CRRA. Let

$$u(c) = \frac{c^{1-\sigma} - 1}{1-\sigma}, \quad \sigma > 0, \quad \sigma \neq 1.$$

Guess a constant saving share out of output:

$$k' = sAk, \quad c = (1-s)Ak.$$

Next period consumption is

$$c_{t+1}(z) = (1-s)AzsAk_t.$$

The Euler equation becomes

$$\begin{aligned} [(1-s)Ak_t]^{-\sigma} &= \beta A [(1-q)[(1-s)A^2sk_t]^{-\sigma} + q\rho[(1-s)A\rho sAk_t]^{-\sigma}] \\ 1 &= \beta A^{1-\sigma} s^{-\sigma} [(1-q) + q\rho^{1-\sigma}]. \end{aligned}$$

Define

$$H(q, \rho, \sigma) \equiv (1-q) + q\rho^{1-\sigma}.$$

Then

$$s = [\beta A^{1-\sigma} H(q, \rho, \sigma)]^{1/\sigma}.$$

An interior solution requires $0 < s < 1$.

Comparative statics.

$$\frac{\partial \log s}{\partial q} = \frac{1}{\sigma} \frac{\rho^{1-\sigma} - 1}{H}.$$

If $\sigma > 1$, then $\rho^{1-\sigma} > 1$ and s rises with disaster probability. If $\sigma < 1$, s falls with disaster probability. Similarly,

$$\frac{\partial \log s}{\partial \rho} = \frac{1}{\sigma} \frac{q(1-\sigma)\rho^{-\sigma}}{H}.$$

A lower ρ raises saving if $\sigma > 1$ and lowers saving if $\sigma < 1$. In the log limit $\sigma = 1$, $H = 1$ and $s = \beta$, so disaster risk does not change the saving share.

Exam tip

The ambiguous sign is not a mistake. Disaster risk has competing forces: a lower expected return discourages saving, while the desire to self-insure bad future states can encourage saving. CRRA curvature determines which force dominates in this homothetic example.

7.3 Worked example 11: skill accumulation with a linear investment rule

Environment. Skill s_t produces labor income ws_t . The agent allocates income between consumption and investment:

$$c_t + i_t = ws_t, \quad s_{t+1} = (1-\delta)s_t + As_t^{1-\gamma}i_t^\gamma.$$

Bellman equation.

$$V(s) = \max_{0 \leq i \leq ws} \left\{ u(ws - i) + \beta V((1 - \delta)s + As^{1-\gamma}i^\gamma) \right\}.$$

FOC and envelope. For an interior investment choice,

$$u'(c) = \beta V'(s') A \gamma s^{1-\gamma} i^{\gamma-1}.$$

The envelope condition is

$$V'(s) = u'(c)w + \beta V'(s') [(1 - \delta) + A(1 - \gamma)s^{-\gamma}i^\gamma].$$

Log closed form. Let $u(c) = \log c$. Guess

$$V(s) = a + B \log s, \quad i(s) = \eta s.$$

Then

$$c = (w - \eta)s, \quad s' = [(1 - \delta) + A\eta^\gamma] s.$$

Coefficient matching gives

$$B = 1 + \beta B \quad \Rightarrow \quad B = \frac{1}{1 - \beta}.$$

The FOC becomes

$$\frac{1}{(w - \eta)s} = \beta \frac{1}{1 - \beta} \frac{1}{[(1 - \delta) + A\eta^\gamma]s} A \gamma \eta^{\gamma-1}.$$

Equivalently, η solves

$$(1 - \beta)(1 - \delta) + A(1 - \beta + \beta\gamma)\eta^\gamma = \beta A \gamma w \eta^{\gamma-1}.$$

Thus investment is linear in skill: $i(s) = \eta s$.

Comparative statics in depreciation. Define

$$F(\eta, \delta) = (1 - \beta)(1 - \delta) + A(1 - \beta + \beta\gamma)\eta^\gamma - \beta A \gamma w \eta^{\gamma-1}.$$

At an interior solution,

$$F_\delta = -(1 - \beta) < 0,$$

and

$$F_\eta = A \gamma \eta^{\gamma-2} [(1 - \beta + \beta\gamma)\eta + \beta w(1 - \gamma)] > 0.$$

Therefore

$$\frac{d\eta}{d\delta} = -\frac{F_\delta}{F_\eta} > 0.$$

Higher depreciation raises the optimal investment flow per unit of skill. Intuitively, when skill erodes faster, replacing skill becomes more valuable at the margin.

7.4 Worked example 12: endogenous population growth and AK technology

Environment. Aggregate output is AK_t . Aggregate feasibility is

$$C_t + K_{t+1} = AK_t.$$

Population evolves according to

$$N_{t+1} = gN_t \left(\frac{C_t}{N_t} \right)^\gamma, \quad \gamma \in (0, 1).$$

Let $c_t = C_t/N_t$ and $k_t = K_t/N_t$.

Per capita law of motion.

$$k_{t+1} = \frac{K_{t+1}}{N_{t+1}} = \frac{N_t(Ak_t - c_t)}{gN_t c_t^\gamma} = \frac{Ak_t - c_t}{gc_t^\gamma}.$$

Bellman equation.

$$V(k) = \max_{0 < c < Ak} \left\{ \log c + \beta V \left(\frac{Ak - c}{gc^\gamma} \right) \right\}.$$

Guess $V(k) = a + b \log k$ and $c = mk$. Substitution shows the coefficient on $\log k$ is

$$1 + \beta b(1 - \gamma),$$

so

$$b = \frac{1}{1 - \beta(1 - \gamma)}.$$

The FOC is

$$\frac{1}{c} + \beta b \left[-\frac{1}{Ak - c} - \frac{\gamma}{c} \right] = 0.$$

Using the value of b yields

$$\boxed{c(k) = (1 - \beta)Ak.}$$

Then

$$k' = \frac{\beta Ak}{g[(1 - \beta)Ak]^\gamma} = \frac{\beta A^{1-\gamma}}{g(1 - \beta)^\gamma} k^{1-\gamma}.$$

The steady-state per capita capital stock solves

$$k^* = \left[\frac{\beta A^{1-\gamma}}{g(1 - \beta)^\gamma} \right]^{1/\gamma}.$$

Long-run population growth is

$$\frac{N'}{N} = gc^\gamma = g[(1 - \beta)Ak^*]^\gamma = \beta A.$$

Thus a higher g lowers long-run per capita consumption enough to leave long-run gross population growth unchanged.

7.5 Worked example 13: project choice with survival probability

Recognition clues. A once-and-for-all discrete project choice plus log utility and linear returns gives conditional value functions that are affine in $\log k$.

Conditional problem. Let $i \in \{S, R\}$ be a project with return A_i and survival probability γ_i . If the project fails, the agent receives constant benefit $b > 0$ forever, with value

$$B_0 = \frac{\log b}{1 - \beta}.$$

Conditional on choosing project i , the active-state Bellman equation is

$$V_i(k) = \max_{0 \leq k' \leq k} \left\{ \log(k - k') + \beta [\gamma_i V_i(A_i k') + (1 - \gamma_i) B_0] \right\}.$$

The initial period has the same form because $c_0 = k_0 - k_1$ and the project starts paying next period.

Guess and policy. Guess $V_i(k) = a_i + b_i \log k$. The FOC is

$$\frac{1}{k - k'} = \frac{\beta \gamma_i b_i}{k'} \Rightarrow k' = \frac{\beta \gamma_i b_i}{1 + \beta \gamma_i b_i} k.$$

Coefficient matching gives

$$b_i = 1 + \beta \gamma_i b_i \Rightarrow b_i = \frac{1}{1 - \beta \gamma_i}.$$

Therefore

$$k' = \beta \gamma_i k, \quad c = (1 - \beta \gamma_i) k.$$

The intercept is

$$a_i = \frac{\log(1 - \beta \gamma_i) + \beta \gamma_i b_i \log(A_i \beta \gamma_i) + \beta(1 - \gamma_i) B_0}{1 - \beta \gamma_i}.$$

Cutoff rule. Since $\gamma_S > \gamma_R$, we have $b_S > b_R$. The safe project is chosen iff

$$a_S + b_S \log k_0 \geq a_R + b_R \log k_0.$$

Thus there is a cutoff

$$\bar{k}_0 = \exp \left(-\frac{a_S - a_R}{b_S - b_R} \right)$$

such that the safe project is chosen iff $k_0 \geq \bar{k}_0$.

Classmate claim. If $A_S > A_R$ and $\gamma_S > \gamma_R$, the classmate might claim that project S leads to higher initial consumption. The policy says the opposite:

$$c_0^i = (1 - \beta \gamma_i) k_0.$$

Since $\gamma_S > \gamma_R$, $c_0^S < c_0^R$. A project that is more likely to survive makes future investment more valuable, so the agent saves more initially.

7.6 Worked example 14: alternating management rights

Independent management. Two identical agents split the asset and each independently solves the AK log problem. Each uses

$$k' = \beta Ak,$$

so aggregate asset stock evolves as

$$K_{t+1} = \beta AK_t.$$

Alternating management. In the joint-ownership case, the current manager chooses current consumption c , private storage s for next period's own consumption, and joint saving k' . The problem is

$$\begin{aligned} V_i(k) = \max_{c, c', s, k' \geq 0} \quad & \log c + \beta \log c' + \beta^2 V_i(g_{-i}(k')) \\ \text{s.t.} \quad & c + s + k' = Ak, \\ & c' = As. \end{aligned}$$

Seek a symmetric linear rule $g_i(k) = g(k) = mAk$ and $V_i(k) = a + b \log k$. Substitute $c' = As$ and $g(k') = mAk'$. The objective is

$$\log c + \beta \log(As) + \beta^2[a + b \log(mAk')].$$

FOCs with multiplier λ on $c + s + k' = Ak$ are

$$\frac{1}{c} = \lambda, \quad \frac{\beta}{s} = \lambda, \quad \frac{\beta^2 b}{k'} = \lambda.$$

Thus

$$s = \beta c, \quad k' = \beta^2 bc.$$

Coefficient matching gives

$$b = 1 + \beta + \beta^2 b \quad \Rightarrow \quad b = \frac{1}{1 - \beta}.$$

Using the resource constraint,

$$c(1 + \beta + \beta^2 b) = Ak.$$

Since $1 + \beta + \beta^2/(1 - \beta) = 1/(1 - \beta)$,

$$c = (1 - \beta)Ak, \quad s = \beta(1 - \beta)Ak, \quad k' = \beta^2 Ak.$$

Therefore aggregate joint asset evolves as

$$K_{t+1} = \beta^2 AK_t.$$

It grows more slowly under alternating management because each manager diverts part of current output into private one-period storage for her own next-period consumption before the other manager takes control.

7.7 Worked example 15: multi-stage seeding, harvesting, and processing

Recognition clues. The problem has a production pipeline. A variable chosen today affects harvest next period, and harvest processed today becomes consumption next period. The state is not just one stock; timing is the whole problem.

Environment. Preferences are

$$\sum_{t=0}^T \beta^t u(c_t), \quad c_t = f(m_t).$$

Seeds s_t planted in period $t - 1$ produce harvest $h(s_t)$ in period t . Harvest in period t can be allocated to seeds s_{t+1} or processing input m_{t+1} :

$$s_{t+1} + m_{t+1} \leq h(s_t), \quad t = 0, \dots, T.$$

Initial s_0 and m_0 are given.

Planner problem.

$$\begin{aligned} \max_{\{s_{t+1}, m_{t+1}\}_{t=0}^T} \quad & \sum_{t=0}^T \beta^t u(f(m_t)) \\ \text{s.t.} \quad & s_{t+1} + m_{t+1} \leq h(s_t), \quad t = 0, \dots, T, \\ & s_{t+1} \geq 0, \quad m_{t+1} \geq 0, \\ & s_0, m_0 \text{ given.} \end{aligned}$$

KKT conditions. Let $\lambda_t \geq 0$ multiply $h(s_t) - s_{t+1} - m_{t+1} \geq 0$. For $t = 1, \dots, T$, the FOC for m_t is

$$\beta^t u'(f(m_t)) f'(m_t) - \lambda_{t-1} + \nu_t = 0,$$

where ν_t is the nonnegativity multiplier for m_t . For $t = 1, \dots, T$, the FOC for s_t is

$$\lambda_t h'(s_t) - \lambda_{t-1} + \mu_t = 0,$$

where μ_t is the nonnegativity multiplier for s_t . Complementary slackness is

$$\begin{aligned} \lambda_t [h(s_t) - s_{t+1} - m_{t+1}] &= 0, \\ \mu_t s_t &= 0, \\ \nu_t m_t &= 0. \end{aligned}$$

Terminal logic. Because m_{T+1} would only produce consumption in period $T + 1$, it has no value in a $0, \dots, T$ problem. Similarly, planting s_T produces harvest at T , but after processing delay it can only affect consumption beyond T . The final useful harvest is allocated to m_T , not to new seeds. Under interior useful consumption, the no-waste terminal implication is that late pipeline variables with no within-horizon payoff are zero.

Euler equation. If $m_t > 0$ and $s_t > 0$ for relevant interior dates,

$$\lambda_{t-1} = \beta^t u'(f(m_t)) f'(m_t), \quad \lambda_{t-1} = \lambda_t h'(s_t).$$

Combining with the m_{t+1} condition gives

$$u'(f(m_t)) f'(m_t) = \beta h'(s_t) u'(f(m_{t+1})) f'(m_{t+1}).$$

This says: one more unit of harvest processed for consumption next period has marginal utility value on the left; one more unit used as seed produces $h'(s_t)$ units of future harvest, processed into next utility value on the right.

Common trap

Pipeline problems require date discipline. If m_t produces c_t , but m_t was chosen from last period's harvest, then m_T is useful and m_{T+1} is not. Do not blindly copy $k_{T+1} = 0$ from the one-stage OGM.

8 Additional formulation variants

8.1 Growth with fixed income

If the agent receives fixed income y in addition to capital income, the recursive problem is

$$V_y(k) = \max_{0 \leq k' \leq f(k)+y} \{u(f(k) + y - k') + \beta V_y(k')\}.$$

The state is still k if y is constant. If income follows $y' = h(y)$, the state is (k, y) :

$$V(k, y) = \max_{0 \leq k' \leq f(k)+y} \{u(f(k) + y - k') + \beta V(k', h(y))\}.$$

If $y' = h(y, y_{-1})$, the state must include the lag: (k, y, y_{-1}) .

8.2 Labor adjustment costs

If labor is chosen and adjustment from n to n' costs $C(n, n')$, current labor is a state:

$$V(k, n) = \max_{c, k', n'} \{u(c) + \beta V(k', n')\}$$

subject to

$$c + k' + C(n, n') \leq f(k, n'), \quad k' \geq 0, \quad n' \in [0, 1].$$

The trap is to omit n from the state even though the adjustment cost depends on it.

8.3 Costly technology adoption

Let technology status be $a \in \{L, H\}$, with $A_H > A_L$. If $a = H$, the planner never returns to L :

$$V_H(k) = \max_{0 \leq k' \leq A_H f(k)} \{u(A_H f(k) - k') + \beta V_H(k')\}.$$

If $a = L$, the planner chooses whether to adopt:

$$V_L(k) = \max \left\{ \max_{0 \leq k' \leq A_L f(k)} [u(A_L f(k) - k') + \beta V_L(k')], \max_{0 \leq k' \leq A_L f(k) - C} [u(A_L f(k) - C - k') + \beta V_H(k')] \right\}.$$

The adoption cost creates an option value of waiting: adopting later may be better when current resources are scarce because the fixed cost has high marginal utility cost.

8.4 Irreversible work choice

A once-and-for-all work decision in period 0 is solved by conditional value functions:

$$V_1(a) = \max_{c, a'} \{u(c) - \phi + \beta V_1(a')\},$$

$$V_0(a) = \max_{c, a'} \{u(c) + \beta V_0(a')\},$$

with budget constraints depending on income y if working and no income if not working. The period-0 choice is work iff $V_1(a_0) \geq V_0(a_0)$. Proving a threshold in a_0 usually requires showing how the difference $V_1(a) - V_0(a)$ moves with assets.

9 Closed-form catalog

Model	Policy/value	Conditions and intuition
Quadratic reaching	$A = (\beta - 2 + \sqrt{\beta^2 + 4})/(2\beta)$, $x' = x/(2 + \beta A)$.	Discounted quadratic LQR-style problem; state converges geometrically to zero.
Cake eating CRRA	$w' = \beta^{1/\sigma} w$, $c = (1 - \beta^{1/\sigma})w$.	Fixed stock; consumption declines because impatience dominates no return.
Finite log-linear OGM	$c_t = (\beta R)^t c_0$, $c_0 = \frac{1-\beta}{1-\beta^{T+1}} R k_0$.	Consumption grows if $\beta R > 1$ and falls if $\beta R < 1$.
Finite log Cobb-Douglas	$s_t = \frac{\alpha\beta[1-(\alpha\beta)^{T-t}]}{1-(\alpha\beta)^{T-t+1}}$.	Saving falls near terminal date; early saving approaches $\alpha\beta$.
Infinite log Cobb-Douglas	$k' = \alpha\beta A k^\alpha$, $c = (1 - \alpha\beta) A k^\alpha$.	Marginal product term lowers saving share relative to AK.
AK log	$k' = \beta A k$, $c = (1 - \beta) A k$.	Saving share out of output is β ; state growth is βA .
Natural resource log/power	$l_t = (1-\beta)[\beta(1+g)]^t x_0$, $c_t = l_t^\gamma$.	Extraction grows, is constant, or falls depending on $\beta(1+g)$.
Disaster AK CRRA	$s = [\beta A^{1-\sigma}((1-q) + q\rho^{1-\sigma})]^{1/\sigma}$.	Disaster risk changes saving through expected marginal utility, not just expected capital.
Skill accumulation log	$i(s) = \eta s$, with η solving the scalar FOC.	Homogeneity in (s, i) makes the policy linear.
Endogenous population AK	$c = (1 - \beta) A k$, $k' = \frac{\beta A^{1-\gamma}}{g(1-\beta)^\gamma} k^{1-\gamma}$.	Higher g lowers per capita k ; long-run population growth remains βA .
Project survival	$V_i(k) = a_i + \frac{1}{1-\beta\gamma_i} \log k$, $k' = \beta\gamma_i k$.	Higher survival raises continuation weight and saving; project comparison yields a cutoff.
Alternating management	$k' = \beta^2 A k$.	Current manager saves less into joint asset because she also privately stores for next-period consumption.

10 Common traps

- Trap 1. Wrong state.** If a variable predicts future payoff, it is a state. If a cost depends on last period's choice, last period's choice is a state.
- Trap 2. Wrong timing.** Accepted jobs may start next period; projects may fail before output; processing may delay consumption.
- Trap 3. Terminal date confusion.** In one-stage finite OGM, $k_{T+1} = 0$. In delayed pipeline models, a variable may become useless earlier.
- Trap 4. Using Euler at a corner.** First check interiority. Inada often gives $c > 0$, but it does not always give $k' > 0$ or extraction interiority.
- Trap 5. Forgetting the envelope.** In recursive problems, the Euler equation usually needs the envelope condition. Do not differentiate $V(g(k))$ and forget that the FOC kills the $g'(k)$ term.
- Trap 6. Matching the wrong coefficient.** In $V = a + b \log k$, match the coefficient on $\log k$ after substituting the optimal policy, not before.
- Trap 7. Mistaking saving share for growth rate.** With $k' = sAk$, s is the share of output saved and sA is the state growth factor.
- Trap 8. Ignoring constants in discrete choice.** In project choice, constants determine the cutoff location even when slopes determine its direction.
- Trap 9. Treating all risk as lowering saving.** Disaster risk can raise saving through precautionary motives when curvature is high.
- Trap 10. No verification.** A guessed policy is not a solution until feasibility, concavity/sufficiency, and transversality or boundedness are checked.

11 Exercise bank

11.1 A. Guess-the-state drills

For each problem, list the minimal state vector and write one Bellman equation. Do not solve.

- A1. **Skill: state choice.** Income alternates deterministically between y_H and y_L . The agent saves in capital with return $f(k)$.
- A2. **Skill: state choice.** Income satisfies $y' = h(y, y_{-1})$.
- A3. **Skill: pipeline timing.** Seeds planted today harvest next period; harvested goods require one period of processing before consumption.
- A4. **Skill: discrete adoption.** Productivity is low until the agent pays fixed cost C ; after adoption productivity is permanently high.
- A5. **Skill: lagged adjustment.** Labor adjustment cost is $C(n, n')$.
- A6. **Skill: stopping.** A machine can be replaced at fixed cost; if not replaced, its productivity follows $z' = h(z)$.
- A7. **Skill: project status.** A risky project can fail; after failure, insurance benefit b is paid forever.
- A8. **Skill: alternating control.** Two agents alternate management rights over a joint asset.

11.2 B. Mechanical derivation drills

- B1. **Skill: KKT.** Derive KKT conditions for finite OGM with $c_t + k_{t+1} \leq f(k_t)$ and $k_{t+1} \geq 0$.
- B2. **Skill: Euler.** Starting from $V(k) = \max_{k'} \{u(f(k) - k') + \beta V(k')\}$, derive FOC, envelope, and Euler.
- B3. **Skill: finite horizon.** For log-linear OGM, derive c_0 using the lifetime resource constraint.
- B4. **Skill: saving recursion.** For finite log Cobb-Douglas OGM, derive $s_t(1 - s_{t+1}) = \alpha\beta(1 - s_t)$.
- B5. **Skill: transversality.** For infinite OGM, explain why Euler alone is not sufficient and state the TC.
- B6. **Skill: stopping.** Show that $TV(x) = \max\{x, \beta V(h(x))\}$ is a contraction.
- B7. **Skill: bounded state.** In stochastic growth with $y = zk^\alpha$ and full depreciation, find a $\bar{k}$ such that the state can be restricted to $[0, \bar{k}]$.
- B8. **Skill: envelope.** In the skill accumulation problem, derive the envelope condition.

11.3 C. Closed-form drills

- C1. **Skill: quadratic guess.** Solve the infinite-horizon reaching-center problem and derive the unique nonnegative A .
- C2. **Skill: CRRA cake eating.** Derive $c(w)$ and $w'(w)$ for CRRA cake eating.
- C3. **Skill: log Cobb-Douglas.** Guess $V = a + b \log k$ and solve for b and the policy.
- C4. **Skill: AK log.** Solve $V(k) = \max_{k'} \{\log(Ak - k') + \beta V(k')\}$.
- C5. **Skill: resource extraction.** With $u(c) = \log c$ and $f(l) = l^\gamma$, derive l_t and c_t .
- C6. **Skill: disaster CRRA.** Derive the disaster-risk saving share and sign comparative statics in q .
- C7. **Skill: human capital.** Show $i(s) = \eta s$ and derive the equation for η .
- C8. **Skill: endogenous population.** Derive $c(k)$ and long-run population growth.
- C9. **Skill: alternating control.** Derive $k' = \beta^2 Ak$.

11.4 D. Past-qual-style creative applications

- D1. **Skill: natural-resource interpretation.** Two agents have resource growth rates $g_2 > g_1$. Without solving equilibrium prices, who wants more date-0 consumption claims and why?
- D2. **Skill: cutoff.** In the project survival model, prove the safe project is chosen above a cutoff.
- D3. **Skill: classmate claim.** If $A_S > A_R$ and $\gamma_S > \gamma_R$, is initial consumption higher under project S ?
- D4. **Skill: recursive monotonicity.** In an OGM with fixed income y , argue how higher y affects the marginal value of capital.
- D5. **Skill: technology adoption.** Explain why an agent may wait to adopt a better technology even when $A_H > A_L$ forever.

- D6. **Skill: disaster intuition.** Give the two forces determining the response of saving to higher q .
- D7. **Skill: management rights.** Compare aggregate growth under independent and alternating management.
- D8. **Skill: pipeline terminal condition.** In the seeding-processing model, identify which late variables are zero and why.

11.5 E. Economist-claim questions

- E1. **Skill: sign check.** “If β rises in finite OGM, consumption must rise at every date.” Evaluate.
- E2. **Skill: policy versus growth.** “In AK log growth, the saving rate is βA .” Evaluate.
- E3. **Skill: disaster.** “A higher disaster probability always lowers saving because the expected return is lower.” Evaluate.
- E4. **Skill: population.** “In endogenous population growth, higher g raises long-run population growth.” Evaluate.
- E5. **Skill: project choice.** “A project with higher return and higher survival always implies higher current consumption.” Evaluate.
- E6. **Skill: stopping.** “A larger tree should always be left to grow because revenue is larger.” Evaluate.

12 Answer sketches

12.1 A. Guess-the-state sketches

- A1. State is (k, j) where $j \in \{H, L\}$ indicates the current income state. Use $V_H(k)$ and $V_L(k)$.
- A2. State is (k, y, y_{-1}) because y' depends on both current and lagged income.
- A3. State includes the pipeline stocks available at the start of the period, e.g. seeds planted for current harvest and produce currently in processing. A sequential KKT formulation is often cleaner.
- A4. State is (k, a) with $a \in \{L, H\}$. V_H has no adoption choice; V_L compares adopt versus wait.
- A5. State is (k, n) because current labor affects the cost of changing to n' .
- A6. State is machine productivity z ; Bellman compares replace value and continue value.
- A7. State is (k, a) with $a \in \{\text{alive}, \text{failed}\}$. Failed state has value $\log b/(1 - \beta)$ under log utility.
- A8. State is (k, i) where i is current manager, or write separate V_1, V_2 with alternating continuation.

12.2 B. Mechanical derivation sketches

- B1. Use multipliers λ_t and μ_t . FOCs imply $u'(c_t) = \lambda_t$ and $-\lambda_t + \mu_t + \beta\lambda_{t+1}f'(k_{t+1}) = 0$. Terminal condition gives $k_{T+1} = 0$.
- B2. FOC: $-u'(f(k) - k') + \beta V'(k') = 0$. Envelope: $V'(k) = u'(f(k) - g(k))f'(k)$. Euler follows by substituting next-period envelope.

- B3. Euler gives $c_t = (\beta R)^t c_0$. PV resource constraint gives $c_0 \sum_{t=0}^T \beta^t = Rk_0$.
- B4. Substitute $c_t = (1 - s_t)Ak_t^\alpha$ and $k_{t+1} = s_t Ak_t^\alpha$ into $1/c_t = \beta \alpha Ak_{t+1}^{\alpha-1}/c_{t+1}$.
- B5. Euler is a second-order difference equation. In infinite horizon, one boundary condition k_0 leaves many paths; TC selects the optimum.
- B6. For any V, W , $|\max\{x, \beta V(h(x))\} - \max\{x, \beta W(h(x))\}| \leq \beta \|V - W\|_\infty$.
- B7. Since labor is one and output is at most $\bar{z}k^\alpha$, choose $\bar{k}$ satisfying $\bar{z}\bar{k}^\alpha \leq \bar{k}$, i.e. $\bar{k} \geq \bar{z}^{1/(1-\alpha)}$.
- B8. Differentiate current payoff and next state with respect to s : $V'(s) = u'(c)w + \beta V'(s')[(1 - \delta) + A(1 - \gamma)s^{-\gamma}i^\gamma]$.

12.3 C. Closed-form sketches

- C1. Guess $v = Ax^2$. Minimized coefficient is $(1 + \beta A)/(2 + \beta A)$. Solve $A = (1 + \beta A)/(2 + \beta A)$.
- C2. Euler gives $c_{t+1} = \beta^{1/\sigma} c_t$. Resource exhaustion gives $c_0 = (1 - \beta^{1/\sigma})w_0$.
- C3. FOC yields $k' = \beta b/(1 + \beta b)Ak^\alpha$. Match $b = \alpha + \alpha\beta b$; policy share is $\alpha\beta$.
- C4. Same as C3 with $\alpha = 1$: $b = 1/(1 - \beta)$ and $k' = \beta Ak$.
- C5. Euler gives $l_{t+1} = \beta(1 + g)l_t$. Resource PV gives $l_0 = (1 - \beta)x_0$.
- C6. Use $k' = sAk$ and CRRA Euler. Get $1 = \beta A^{1-\sigma} s^{-\sigma} [(1 - q) + q\rho^{1-\sigma}]$.
- C7. Guess $V = a + B \log s$ and $i = \eta s$. Match $B = 1/(1 - \beta)$ and use the FOC to get the scalar equation for η .
- C8. Bellman is $V(k) = \max_c \{\log c + \beta V((Ak - c)/(gc^\gamma))\}$. Guess $a + b \log k$; FOC gives $c = (1 - \beta)Ak$.
- C9. FOCs imply $s = \beta c$, $k' = \beta^2 bc$. Match $b = 1 + \beta + \beta^2 b = 1/(1 - \beta)$, so $k' = \beta^2 Ak$.

12.4 D. Past-qual-style sketches

- D1. Agent 2 has a faster-growing resource and therefore a steeper future endowment profile. Agent 2 is relatively more willing to sell date-0 consumption claims and buy future claims; agent 1 wants more date-0 claims.
- D2. Conditional values are $a_i + b_i \log k$ with $b_i = 1/(1 - \beta\gamma_i)$. Since $\gamma_S > \gamma_R$, $b_S > b_R$, so the difference crosses once in $\log k$.
- D3. No. Conditional policy is $c_0^i = (1 - \beta\gamma_i)k_0$. Higher survival raises saving and lowers initial consumption.
- D4. Higher noncapital income reduces the marginal utility value of an extra unit of capital through a wealth effect under concavity; consumption rises and saving tends to fall in the standard normal case.
- D5. Fixed adoption cost is paid when current marginal utility may be high. Waiting lets the agent build resources before paying the fixed cost, so option value can justify delay.
- D6. A higher q lowers expected physical survival of capital but raises the marginal utility of resources in bad future states. The sign depends on curvature in the CRRA AK model.

- D7. Independent management gives $K' = \beta AK$. Alternating management gives $K' = \beta^2 AK$. Alternating grows slower because $\beta^2 < \beta$.
- D8. Any variable that only yields post- T consumption or post- T harvest has no terminal value; set it to zero unless the problem gives a salvage value.

12.5 E. Economist-claim sketches

- E1. False in general. A higher β tilts consumption toward the future; early consumption can fall while later consumption rises.
- E2. False. The saving share out of output is β . The state growth factor is βA .
- E3. False. With CRRA, s rises with q if $\sigma > 1$, falls if $\sigma < 1$, and is unchanged in the log case.
- E4. False in the endogenous-population AK model. Long-run gross population growth is βA , independent of g .
- E5. False. In the survival project model, higher survival raises saving, so current consumption is lower.
- E6. False. If discounted growth is weak above the threshold, cutting is optimal even though the tree is large.

13 Past-qual and problem-set practice map

Source	Problem archetype	What to practice from this module
PS1 2026, Reaching center	Quadratic dynamic problem	Section 4.1; finite-horizon induction and infinite-horizon coefficient fixed point.
PS1 2026, Tree cutting	Stopping problem	Section 4.2; Bellman max, contraction, cutoff proof.
PS1 2026, Growth with leisure	Abstract DP mapping	State/control/feasible correspondence; monotonicity and concavity proof logic.
PS1 2026, Stochastic growth	Bounded state and stochastic Bellman	State restriction and value-function properties.
HW1 Galina	Multi-stage seeding and finite OGM	Sections 3.1 and 6.7; KKT and terminal timing.
HW2 Galina	Finite log-linear and log Cobb-Douglas OGM	Sections 5.1–5.2; saving-rate recursions.
HW3 Galina	Cake eating and recursive formulation	Section 4.3; state selection with exogenous income.
HW4 Galina	Endogenous population and skill accumulation	Sections 6.3 and 6.4; recursive comparative statics.
HW5 Galina	Disaster risk and recursive setup variants	Section 6.2; labor adjustment and costly adoption variants.

Source	Problem archetype	What to practice from this module
2018 Qual	Natural resource/lumber extraction	Section 6.1; resource Euler and closed-form extraction.
2019 Qual	Irreversible work decision	Conditional value functions and threshold logic.
2020 Qual	Growth with public leisure	Recursive formulation; FOC/envelope; steady-state intuition.
2021 Qual	Skill accumulation	Section 6.3; $i(s) = \eta s$ and comparative statics in depreciation.
2022 Qual	Project choice with survival	Section 6.5; conditional value functions and cutoff.
2023 Qual	Alternating management rights	Section 6.6; symmetric linear rules and growth comparison.
2024 Qual	Disaster-risk AK	Section 6.2; stochastic Euler and CRRA saving share.

14 Quality audit

Verification checklist

1. Source inventory is explicit and includes extraction caveats.
2. Blueprint precedes the full packet.
3. Every canonical dynamic model has states, choices, constraints, Bellman/sequential formulation, FOC/envelope/Euler or cutoff, intuition, and traps.
4. Closed forms are derived through guesses and coefficient matching rather than merely stated.
5. Current PS1 and relevant past quals are mapped to practice sections.
6. Exercises cover state selection, recursive formulation, coefficient matching, closed forms, cutoff arguments, and Economist X claims.
7. Notation is consistent: k' is next capital in recursive problems; s is a saving share only where explicitly defined; i is investment in skill/project contexts.
8. Remaining limitation: lecture-note PDFs were handwritten scans, so some note details were reconstructed using homeworks and past quals when OCR was unclear.

15 Final study checklist

Before attempting old quals, make sure you can do the following without notes:

1. Write a Bellman equation with the correct state in under two minutes.
2. Derive FOC, envelope, and Euler from a recursive OGM.
3. Explain the terminal condition in a finite-horizon problem.

4. Recognize when $V = a + b \log k$, $V = Bk^{1-\sigma}/(1-\sigma)$, or $V = Ax^2$ is the correct guess.
5. Match coefficients cleanly and solve for the policy share.
6. Check feasibility and interiority of the policy.
7. State the transversality condition for an infinite-horizon sequential problem.
8. Solve cake eating, AK log, and log Cobb-Douglas growth from memory.
9. Derive the natural-resource Euler equation and closed-form extraction path.
10. Derive the disaster-risk CRRA saving share and interpret its comparative statics.
11. Show why skill investment is linear in skill under log utility.
12. Prove a cutoff by comparing conditional value functions.
13. Translate every Economist X claim into a sign or inequality and test it against a policy formula.